

# Sheng Luo PhD <sup>1</sup>

## Contact

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## Education

Ph.D. in Biostatistics, Department of Biostatistics, Johns Hopkins University, 2008.  
Dissertation: "Mixed Effects Stochastic Process Models of Smoking Cessation Behavior."  
Dissertation advisor: Ciprian M. Crainiceanu. Co-advisor: Thomas A. Louis.

M.S. in Statistics/Mathematics, Department of Mathematics, University of Texas at Arlington, 2003.

M.S. in Mechanical Engineering, Huazhong University of Science and Technology, China, 2000.

B.S. in Mechanical Engineering, Huazhong University of Science and Technology, China, 1996.

## Professional experience

Full Professor, Department of Biostatistics and Bioinformatics, Duke University, September 2019  
- Present

Associate Professor, Department of Biostatistics and Bioinformatics, Duke University, August 16, 2017 - August 2019

Tenured Associate Professor, Department of Biostatistics, School of Public Health, University of Texas Health Science Center at Houston, September 2015 - August 15, 2017

Assistant Professor, Department of Biostatistics, School of Public Health, University of Texas Health Science Center at Houston, July 2008-August 2015

Predoctoral fellow and statistical consultant, Biostatistics Branch, Division of Cancer Epidemiology and Genetics, National Cancer Institute, National Institutes of Health, 2005-2008, Mentor: Dr. Nilanjan Chatterjee

Research assistant, Department of Molecular Microbiology and Immunology, Johns Hopkins University, 2004-2005

Research assistant, Department of Environmental Health Sciences, Johns Hopkins University, 2004-2005

Research assistant, Department of Mathematics, University of Texas at Arlington, 2000-2003

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<sup>1</sup>Last updated April 20, 2026

## Honors and awards

1. **Fellow, American Statistical Association (2020)**, citation reads: "For excellent contributions to statistical methods in longitudinal and survival analysis; for broad scientific contributions to the field of neurological diseases; for exemplary mentorship of students and significant service to the profession. "
2. **Elected Member, International Statistical Institute (2020)**
3. Young Investigator Award, the American Statistical Association, Statistics in Epidemiology Section, 2007
4. Laha Award, the Institute of Mathematical Statistics, 2007
5. Fellow Award for Research Excellence, the National Institutes of Health (NIH), 2007
6. Distinguished Student Paper Award, the International Biometrics Society, Eastern North American Region (ENAR), 2007
7. Overall first-place winner of the Scientific Poster Competition, the Delta Omega Public Health Honor Society, Alpha Chapter, Johns Hopkins Bloomberg School of Public Health, 2007
8. Louis I. and Thomas D. Dublin Award for the Advancement of Epidemiology and Biostatistics, Johns Hopkins Bloomberg School of Public Health, 2007
9. Cancer Training Research Award, the National Cancer Institute, NIH, 2005–2008

## Grants and contract awards

### Research Support

#### Ongoing research support

1. NIH/NIA, R01AG064803, "Integrative modeling and dynamic prediction of Alzheimer's disease", 9/15/2020 - 5/31/2026, role: **PI**.
2. NIH/NIA, P30AG072958, "Duke/UNC Alzheimer's Disease Research Center (ADRC)", 9/1/2021-6/30/2026, MPIs: Heather Whitson, Gwenn Garden (UNC), role: **Co-PI** of Data Management and Statistics Core.
3. NIH/NIDDK, U01DK140923, "The Bidirectional Gut-Microbiota-Brain Axis in Parkinson's disease: integrating mechanistic biomarkers of disease severity and progression", 8/15/2024-6/30/2029, MPIs: Ali Keshavarzian (Rush), Christopher G. Goetz (Rush), and Amol Sharma (Augusta), role: **PI** of Duke sub-contract.
4. NIH/NCI, R01CA297227, "AI for Differentiation of Low vs High Risk Thyroid Nodules on Ultrasound", 9/1/2025-8/31/2029, MPIs: Ben Wildman-Tobriner, Maciej Mazurowski, Miriam Morey, role: co-investigator.
5. NIH/NCBIB, P41EB028744, "Center for Virtual Imaging Trials", 4/1/2021-12/31/2026, PI: Ehsan Samei, role: co-investigator.
6. NIH/NIA, R01AG080048, "Common mechanistic biomarkers of vascular and neuro-degeneration", 1/1/2023-12/31/2027, PI: Yongmei Liu, role: co-investigator.
7. NIH/NIA, R01AG081322, "Metabolic age to define influences of the lipidome on brain aging in Alzheimer's disease", 5/15/2023-3/31/2028, MPI: Rima Kaddurah-Daouk, Matthias Arnold, Peter Meikle, role: co-investigator.
8. NIH/NIA, R01AG084098, "Automated high-purity exosome isolation-based AD diagnostics system (AHEADx)", 9/1/2023-5/31/2028, PI: Tony Jun Huang, role: co-investigator.

9. NIH/NIA, R01AG082335, "Biopsy-based single cell assays for olfactory sensory changes in an Alzheimer's disease cohort", 2/1/2024-11/30/2028, PI: Bradley Goldstein, role: co-investigator.
10. NIH/NCI, R01CA261457, "Computer-aided triage of body CT scans with deep learning", 8/16/2023-7/31/2027, PI: Joseph Lo, role: co-investigator.
11. NIH/NHLBI, R01HL155293, "Accuracy and precision in CT quantification of COPD through virtual imaging trials", 7/1/2021-6/30/2026, PI: Ehsan Abadi, role: co-investigator.
12. NIH/NIBIB, R01EB031575, "Harmonization of breast MRI data", 9/15/2022-6/30/2026, PI: Maciej J Mazurowski, role: co-investigator.
13. NIH, U19NS115388, "Determinants of Incident Stroke Cognitive Outcomes and Vascular Effects on RecoverY", 9/19/2019 - 8/31/2026, PI: Rebecca Gottesman (John Hopkins University), MD, role: co-investigator.

#### **Completed research support**

14. NIH/NIA, R56AG076622, "Trajectories of blood-based biomarkers of AD, their determinants, and ability to predict cognitive impairment", 9/15/2022-8/31/2024, MPIs: Yongmei Liu, Sheng Luo, role: **MPI**.
15. Parkinson's Foundation, award ID: 858345, "Parkinson's Outcomes Project - Statistical Analysis Core", 3/1/2021- 10/31/2023, role: **PI**.
16. CHDI Foundation, project ID: A-16209, "Novel latent trait models of Huntington's disease", 7/1/2021- 12/31/2022, role: **PI**.
17. NIH/NINDS, R01NS091307, "Statistical methods for clinical trials with multivariate longitudinal outcomes", 9/30/2015 - 12/31/2020, role: **PI**.
18. NIH/NIA, R56AG064803, "Integrative modeling and dynamic prediction of Alzheimer's disease", 9/1/2019 - 8/31/2020, role: **PI**.
19. NIH/NIA, R56AG062302, "Shared genetic, epigenetic, and transcriptomic profiles between AD and PTSD: molecular insights into the heterogeneity of neuropsychiatric symptoms in Alzheimers Disease", 9/30/2018 - 8/31/2020, role: **co-PI**.
20. NIH/NICHD, 1R01HD096331, "Vestibulodynia: Understanding Pathophysiology and Determining Appropriate Treatments (Vestibulodynia: UPDATE)", 9/11/2018 - 5/31/2023, PI: Andrea Nackley, PhD, role: co-investigator.
21. NIH/NICHD, 1P50HD093074, "Co-occurring ADHD in young children with ASD: Precursors, detection, neural signatures, and early treatment", 9/7/2017 - 7/31/2022, PI: Geraldine Dawson, PhD, role: faculty statistician.
22. CHDI Foundation, "Dynamic monitoring and risk prediction of Huntington's disease", 7/1/2019-6/30/2020, role: **PI**.
23. NIH/NIDA, R34DA050267, "HEAL Consortium: Establishing innovative approaches for the Healthy Brain and Child Development Study", 9/30/2019 - 3/31/2021, PI: Brian Smith, MD, role: co-investigator.
24. AKili, Inc, "Software Treatment for Actively Reducing Severity of ADHD as Adjunctive Treatment to Stimulant (STARS-ADHD-Adjunctive)", PI: Daniel Laskowitz, MD, role: faculty statistician.
25. KemPharm, Inc, "A multicenter, dose-optimized, double-blind, randomized, placebo-controlled, parallel efficacy laboratory classroom study with KP415 in children with attention-deficit/hyperactivity disorder (ADHD)", PI: Scott H. Kollins, PhD, role: faculty statistician.
26. NIH/NICHD, 5U01HD073984, "Study of oxytocin in autism to improve reciprocal social behaviors (SOARS-B)", 9/4/2012 - 5/31/2028, PI: Linmarie Sikich, MD, role: faculty statistician.

27. NIH/NIMH, "A phase 2a study to evaluate the Kappa Opioid Receptor as a target for the treatment of mood and anxiety spectrum disorders by evaluation of whether LY2456302 engages key neural circuitry related to the hedonic response (FAST-MAS KOR 2a)", PI: Andrew D. Krystal, MD, role: faculty statistician.
28. Intra-Cellular Therapies, Inc, "A randomized, placebo-controlled, double-blind study of safety, tolerability, pharmacokinetics and pharmacodynamics of multiple doses of ITI-214 in patients with idiopathic Parkinson's disease", PI: Laurie Sanders, PhD, role: faculty statistician.
29. CHDI Foundation (a privately-funded, not-for-profit biomedical research organization devoted to Huntington's disease), "FuRST 2.0 (Functional Rating Scale Taskforce) Development and Testing", 15% effort, project period: 2/1/2015-12/31/2017, role: **PI**.
30. International Parkinson and Movement Disorder Society (a professional society representing Parkinson's disease, related neurodegenerative and neurodevelopmental disorders, hyperkinetic movement disorders, and abnormalities in muscle tone and motor control), "Validating the Uniform Parkinson's Disease Rating Scale", 12/15/2012 - 11/30/2017, role: **PI**.
31. International Parkinson and Movement Disorder Society, "Missing values and differential item functioning analysis of the Unified Dyskinesia Rating Scale (UDysRS)", 1/1/2017 - 12/31/2017, role: **PI**.
32. NIH/NINDS, 5U01NS043127, "Parkinson's Disease Clinical Trial: Statistical Center", 10% effort, 06/01/2016 - 05/31/2017, role: **PI**, took over from Dr. Barbara Tilley since June 1, 2016.
33. NIH/NCATS, 5KL2TR000370, "Innovative Statistical Methods for Analyzing Parkinson's Disease Study Data", 75% effort (6/1/2013 - 9/29/2015), 50% (9/30/2015 - 5/31/2016), role: **PI**.
34. National Parkinson Foundation, "Determining the Effects of Early Exercise and Physical Therapy on Outcomes Using the Parkinson's Outcomes Project Dataset", project period: 6/15/2015-6/14/2016, role: **PI**.
35. International Parkinson and Movement Disorder Society, "The use of the MDS-UPDRS validation and translation datasets to investigate DIF and missing values", 3/1/2014 - 2/28/2015, role: **PI**.
36. Dystonia Coalition/Office of Rare Diseases Research (ORDR)/NINDS, "Item Response Theory: An Individualized Approach to Evaluating a Composite Cervical Dystonia Rating Scale", 7/9/2012-12/31/2013, role: consultant.
37. NIH-NINDS, 5U01NS043127, P.I. Barbara Tilley, "Parkinson's Disease Clinical Trial: Statistical Center", 40% effort, 09/01/2009 - 05/31/2013, role: Co-Investigator.
38. Subcontracted from Baylor College of Medicine, through a NIH P30 core grant, PI: Janet Butel, "Baylor-UTHouston Center for AIDS Research", 5% effort, 8/10/2010 - 5/31/2013, role: Co-Investigator.
39. NIH-NHLBI, RC2HL102419-0110, PI: Eric Boerwinkle, "Building on GWAS for NHLBI-diseases: the CHARGE consortium", 10% effort, 11/17/2009-7/31/2012, role: Co-Investigator.
40. UTHSC-SPH, PRIME award, "Classifying Ipsilateral Breast Tumor Relapse: a Novel Bayesian Approach", 09/01/2009 - 08/31/2010, role: **PI**.
41. NIH-NEI, supplement to 5U10EY012471, PI: Robert Hardy, "A Hybrid Method in Combining Treatment Effects from Matched and Unmatched Studies", 25% effort, 10/1/2009 - 9/29/2010, role: Co-Investigator.

## Professional Activities

### Service to Professional Society/Foundation

1. International Parkinson and Movement Disorder Society (MDS), Clinical Outcome Assessments (COA) Translation Steering Committee, **Chair**, Oct, 2021 - Oct 2025. Duties: Recommend and approve non-English translations of the MDS-owned scales MDS-UPDRS, MDS-NMS, and UDysRS; Assist in the development of non-English translation teams, monitor translation development and oversee all aspects of the statistical analysis at each phase of development; Make recommendations for best practices in the translation and validation of digital tools acquired by the Society; Manage use of generated rating scales data owned by MDS.
2. MDS COA Statistical Lead, Oct 2025 - present.
3. National Vaccine Injury Compensation Program (VICP), Health Resources and Services Administration (HRSA), Department of Health and Human Services (HHS), **Medical Panel Expert**, 2021 - present. Duties: review medical records and relevant scientific literature, prepare written reports, and serve as government expert witnesses in hearings before the Court.
4. National Alzheimer's Coordinating Center (NACC), **Data Core Steering Committee**, Fall 2022 - Fall 2025. 2-year term as member and 1-year term as Chair.
5. National Parkinson Study Group (PSG), **Scientific Review Committee**, Nov 1, 2022 - Oct 31, 2025, 3-year term.
6. Huntington Study Group (HSG), **Scientific Review Committee**, Jan, 2023 - present.
7. IQVIA Data Safety Monitoring Board, 2021 - present.
8. National Institute on Aging (NIA) **Biobank Scientific Review Committee**, Oct 2020 - present.
9. American Statistical Association, Biometrics Section Chair (January 2019 - December 2019).
10. American Statistical Association, Committee on Career Development, January 2018 - December 2019.
11. MDS Rating Scales Translation Committee, Chair, June 30, 2017 - September 30, 2021
12. The International Parkinson and Movement Disorder Society (MDS) Rating Scales Steering Committee, vice-chair (September 26, 2019 - September 30, 2021); member (June 30, 2017 - September 25, 2019)
13. MDS Scales Development Committee, June 15, 2013 - June 30, 2017
14. The Parkinson Study Group (PSG) Scientific Review Committee (SRC), December 2013 - January 2017
15. Eastern North American Region (ENAR) of the International Biometric Society (IBS) Regional Advisory Board (RAB), January 1, 2015 - December 31, 2017.
16. International Chinese Statistical Association (ICSA), Biometrics Section Chair, January 2016-December 2016.
17. American Statistical Association, Biometrics Section Publication Officer, January 2015 - December 2016.

### **Grant Review Services**

1. NIH reviewer on Special Emphasis Panel/Scientific Review Group ZRG1 HSS-R (90)S. Review Date: Jan 16, 2026, SRO: Rhoda Keese Moise.
2. NIH reviewer on Special Emphasis Panel /Scientific Review Group ZAG1-Zu-5(M2)B. Review Date: Apr 15-16, 2025, SRO: Janetta Lun.
3. **NIH NMBH (Neurological, Mental and Behavioral Health) Study Section Chartered Member, 07/01/2022-06/30/2024.** Grant reviewed on 10/17 - 10/18/2022, 2/27/2023-2/28/2023, 6/29/2023-6/30/2023, 2/21/2024-2/22/2024, 5/30/2024-5/31/2024.
4. **NIH NAME (Neurological, Aging and Musculoskeletal Epidemiology) Study Section Chartered Member, 07/01/2020-06/30/2022.** Grant reviewed on 10/14 - 10/16/2020; 02/24 - 02/26/2021; 07/07 - 07/09/2021, 10/06 - 10/07/2021, 02/09 - 02/10/2022, 06/08 - 06/09/2022.
5. The Parkinson Study Group (PSG) grant review (review Date: 05/05/2014, 05/22/2015, 07/07/2015, 05/2015, 11/2015, 07/22/2016, 01/27/2017)
6. NIH reviewer on Special Emphasis Panel /Scientific Review Group 2015/10 ZHL1 CSR-D (O1). Review Date: July 29, 2015, SRO: Keary Cope.
7. NIH reviewer on National Heart, Lung, And Blood Institute Clinical Trials Review Committee, CLTR-MA, Review Date: March 3-4, 2016, SRO: Keary Cope.
8. NIH reviewer on Risk Prevention and Health Behavior Special Emphasis Panel, ZRG1-RPHB-C-11. Review Date: November 17-18, 2016, SRO: Claire Gutkin.
9. NIH reviewer on NINDS Special Emphasis Panel, ZNS1-SRB-C-04. Review Date: November 28, 2016, SRO: Tiziana Cogliati.
10. NIH reviewer on Healthcare Delivery and Methodologies (HDM-11) Small Business: Health Informatics Study Section, ZRG1-HDM-G-11. Review Date: March 13, 2017, SRO: Yvonne Ferguson.
11. NIH reviewer on Healthcare Delivery and Methodologies (HDM-11) Small Business: Health Informatics Study Section, ZRG1-HDM-A-11. Review Date: June 26, 2017, SRO: Michael Bloom.
12. NIH reviewer on NINDS Clinical Trial Readiness meeting. Review Date: July 14, 2017, SRO: Ana Olariu.
13. NIH reviewer on Small Business: Health Informatics Special Emphasis Panel, ZRG1-HDM-Z-11. Review Date: November 03, 2017, SRO: Peter Kozel.
14. NIH reviewer on Small Business: Health Informatics Special Emphasis Panel, ZRG1-HDM-H-11. Review Date: March 26-27, 2018, SRO: Sudha Veeraghavan.
15. NIH reviewer on Biobehavioral Mechanisms of Emotion, Stress and Health Study Section, MESH, Review Date: June 7-8, 2018, SRO: Maribeth Champoux.
16. NIH reviewer on Health Decisions for Older People Study Section, National Institute of Aging, Review Date: September 12, 2018, SRO: Carmen Moten.
17. NIH reviewer on Neurological, Aging and Musculoskeletal Epidemiology (NAME) Study Section, Review Date: October 11-12, 2018, SRO: Heidi B. Friedman.
18. NIH reviewer on NIH/CSR Anonymization Study (HDM), Review period: March 2019, SRO: Jeremy Braithwaite.

19. NIH reviewer on Special Emphasis Panel, ZRG1 PSE-A (56) R, Review date: July 18-19, 2019, SRO: Delia Olufokunbi Sam.
20. NIH reviewer on Neurological, Aging and Musculoskeletal Epidemiology (NAME) Study Section, Review Date: October 17-18, 2019, SRO: Heidi B. Friedman.
21. NIH reviewer on Special Emphasis Panel, ZAG1ZIJ-G(M3), Review Date: Feb 25, 2022, SRO: Maurizio Grimaldi.

### **Data and Safety Monitoring Board Services**

1. SEGA - SEdation versus General Anesthesia for Endovascular Therapy in Acute Ischemic Stroke a Randomized Comparative Effectiveness Trial. ClinicalTrials.gov Identifier: NCT03263117, PIs: Drs. Peng Roc Chen, Andrew Barreto, Carlos A. Artime, McGovern Medical School at The University of Texas Health Science Center at Houston, status: completed.
2. An Open-Label Phase I/II Dose Finding Study to Evaluate the Safety, Tolerability, and Preliminary Efficacy of Striatal Administration of BV-101 in Adult Subjects with Early Manifest Huntington's Disease (HD). Regulatory Agency Identifier Number: EudraCT: 2021-003052-16. Sponsor: Asklepios BioPharmaceutical. Status: completed.

### **Steering Committee Services for Clinical Trials**

1. A Phase 2, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy, Safety, Tolerability, Pharmacodynamics, and Pharmacokinetics of BIA 28-6156 in Subjects With Parkinson's Disease With a Pathogenic Variant in the Glucocerebrosidase (GBA1) Gene. Sponsor: BIAL R& D Investments S.A. Status: ongoing.

### **Editorial Boards**

1. Associate Editor, BMC Medical Research Methodology, February 2016 - present, manuscripts handled (22).
2. Associate Editor, Journal of Alzheimer's Disease, January 2018 - present.
3. Editorial board member, Cancer Medicine, June 2018 - present.

**Journal Referee (Number in parenthesis indicates # of manuscripts reviewed):** American Journal of Epidemiology (4), Alzheimer's & Dementia (3), Annals of Applied Statistics (3), Annals of Epidemiology (1), Annals of Human Genetics (1), Assessment (1), Australian & New Zealand Journal of Statistics (1), Bayesian Analysis (3), Biometrical Journal (1), Biometrics (14), Biostatistics (7), BMC Health Services Research (1), BMC Medical Informatics and Decision Making (1), BMC Medical Research Methodology (6), BMC Medicine (1), BMC Neurology (1), BMJ Open (1), Canadian Journal of Statistics (1), Cancer Cell (9), Cell (1), Computational Statistics & Data Analysis (6), Computer Methods and Programs in Biomedicine (1), Communications in Statistics-Theory and Methods (1), Computer Methods and Programs in Biomedicine (1), Journal of Alzheimers Disease (2), Journal of American Statistical Association (6), Journal of Applied Probability and Statistics (1), Journal of Applied Statistics (5), Journal of Clinical Oncology (8), Journal of Consulting and Clinical Psychology (4), Journal of the Royal Statistical Society-Series C: Applied Statistics (2), Journal of Probability and Statistics (1), Journal of Statistical Computation and Simulation (1), Metron (1), Movement Disorders (13), Movement Disorders Clinical Practice (1), Nature (1), Nature Biotechnology (4), Nature

Communications (9), Nature Chemical Biology (1), Nature International Journal of Obesity (1), Nature Medicine (5), Plos One (2), Psychology, Health & Medicine (1), Scientific Reports (1), Statistics in Biopharmaceutical Research (1), Statistics in Medicine (17), Statistical Methods in Medical Research (5), WIREs Computational Statistics (1), Reviewer of International Society for Pharmacoeconomics and Outcomes Research (ISPOR) Assessing Prospective Observational Studies Task Force Draft Report (1).

## Publications

\* represents student/postdoc author under Sheng Luo's supervision; \*\* represents student author of which Sheng Luo serves as committee member; † represents Sheng Luo as corresponding author.

### Statistical methodology papers

1. **Luo S**<sup>†</sup>, Crainiceanu CM, Louis TA, Chatterjee N. (2008). Analysis of smoking cessation patterns using a stochastic mixed-effects model with a latent cured state. *Journal of The American Statistical Association*, 103(483), 1002-13, PMID: 19305513, PMCID: PMC2658598.
2. **Luo S**<sup>†</sup>, Crainiceanu CM, Louis TA, Chatterjee N. (2009). Bayesian inference for smoking cessation with a latent cure state. *Biometrics*, 65(3), 970-8, PMID: 19173701, PMCID: PMC3856570.
3. **Luo S**<sup>†</sup>, Mukherjee B, Chen J, Chatterjee N. (2009). Shrinkage estimation for robust and efficient screening of single SNP association from case-control genome-wide association studies. *Genetic Epidemiology*, 33(8), 740-50, PMID: 19434716, PMCID: PMC3103068.
4. Chatterjee N, Chen YH, **Luo S**, Carroll RJ. (2009). Analysis of case-control association study: SNPs, imputation and haplotypes, *Statistical Science*, 24(4), 489-502, PMID: 20543902, PMCID: PMC2883271.
5. Crainiceanu CM, Caffo B, **Luo S**, Zippunikov V, Punjabi NM. (2011). Population value decomposition, a framework for the analysis of image populations. *Journal of the American Statistical Association*, discussion paper, 106(495), 775-90, PMID: 24415813, PMCID: PMC3886284.
6. Crainiceanu CM, Caffo B, **Luo S**, Zippunikov V, Punjabi NM. (2011). Rejoinder to comments on "Population value decomposition, a framework for the analysis of image populations". *Journal of the American Statistical Association*, 106(495), 803-6.
7. **Luo S**<sup>†</sup>, Yi M\*, Huang X, Hunt KK. (2013). A Bayesian model for misclassified binary outcomes and correlated survival data with applications to breast cancer. *Statistics in Medicine*, 32(13), 2320-34, PMID: 22996169, PMCID: PMC3897718.
8. **Luo S**<sup>†</sup>, Ma J\*, Kiebertz KD. (2013). Robust Bayesian inference for multivariate longitudinal data using normal/independent distributions, *Statistics in Medicine*, 32(22), 3812-28, PMID: 23494809, PMCID: PMC3884581.
9. Chen Y, **Luo S**, Chu H, Wei P. (2013). Bayesian inference on risk differences: an application to multivariate meta-analysis of adverse events in clinical trials, *Statistics in Biopharmaceutical Research*, 5(2), 142-50, PMID: 23853700, PMCID: PMC3706106.
10. Byun J\*\*, Lai D, **Luo, S**, Risser J, Tung B, Hardy RJ. (2013). A hybrid method in combining treatment effects from matched and unmatched studies, *Statistics in Medicine*, 32(28), 4924-37, PMID: 23839782, PMCID: PMC3887129.
11. **Luo S**<sup>†</sup>. (2013). Joint analysis of stochastic processes with application to smoking patterns and insomnia, *Statistics in Medicine*, 32(29), 5133-44, PMID: 23913574, PMCID: PMC3856619.
12. **Luo S**<sup>†</sup>, Chen Y, Su X\*, Chu H. (2014). mmeta: An R package for multivariate meta-analysis, *Journal of Statistical Software*, 56(11), 1-26, PMID: 24904241, PMCID: PMC4043353.



13. **Luo S<sup>†</sup>**. (2014). A Bayesian approach to joint analysis of multivariate longitudinal data and parametric accelerated failure time, *Statistics in Medicine*, 33(4), 580-94, PMID: 24009073, PMCID: PMC3947121.
14. **Luo S<sup>†</sup>**, Su X\*, DeSantis S, Huang X, Yi M\*, Hunt KK. (2014). Joint model for a diagnostic test without a gold standard in the presence of a dependent terminal event, *Statistics in Medicine*, 33(15), 2554-66, PMID: 24473943, PMCID: PMC4209250.
15. Chen Y, **Luo S**, Chu H, Su X\*, Nie L. (2014). An empirical Bayesian method for multivariate meta-analysis with an application in clinical trials. *Communication in Statistics-Theory and Methods*, 43(16), 3536-51, PMID: 25089070, PMCID: PMC4115294.
16. **Luo S<sup>†</sup>**, Wang J\*. (2014). Bayesian hierarchical model for multiple repeated measures and survival Data: an application to Parkinson's disease, *Statistics in Medicine*, 33(24), 4279-91, PMID: 24935619, PMCID: PMC4184935.
17. **Luo S<sup>†</sup>**, Su X\*, Yi M\*, Hunt KK. (2015). Simultaneous inference of a misclassified outcome and competing risks failure time data, *Journal of Applied Statistics*, 42(5), 1080-90, PMID: 25821331, PMCID: PMC4372162.
18. Chen Y, Chu H, **Luo S**, Nie L, Chen S. (2015). Bayesian analysis on meta-analysis of case-control studies accounting for within study correlation. *Statistical Methods in Medical Research*, 24(6), 836-55, PMID: 22143403, PMCID: PMC3683108.
19. Wang J\*, **Luo S<sup>†</sup>**. (2016). Augmented Beta rectangular regression models: A Bayesian perspective, *Biometrical Journal*, 58(1), 206-21, PMID: 26289406, PMCID: PMC5064841 [**\*\* An earlier version won Jue Wang a Student Travel Award from Parkinson's Disease Foundation in year 2014. \*\***]
20. **Luo S<sup>†</sup>**, Chan W, Detry M\*\*, Massman PJ, Doody RS. (2016). Binomial regression with a misclassified covariate and outcome, *Statistical Methods in Medical Research*, 25(1), 101-17, PMID: 22421539, PMCID: PMC3883897.
21. **Luo S<sup>†</sup>**, Lawson AB, He B\*, Elm J, Tilley BC. (2016). Bayesian multiple imputation for missing multivariate longitudinal data from a Parkinson's disease clinical trial, *Statistical Methods in Medical Research*, 25(2), 821-37, PMID: 23242384, PMCID: PMC3883900.
22. Benoit J\*\*, Chan W, **Luo S**, Yeh HW, Doody R. (2016). A hidden Markov model approach to analyze longitudinal ternary outcomes when some observed states are possibly misclassified, *Statistics in Medicine*, 35(9), 1549-57, PMID: 26782946, PMCID: PMC4821697.
23. Chen G\*, **Luo S<sup>†</sup>**. (2016). Robust Bayesian hierarchical model using normal/independent distributions, *Biometrical Journal*, 58(4), 831-51, PMID: 26711558, PMCID: PMC5064853.
24. He B\*, **Luo S<sup>†</sup>**. (2016). Joint modeling of multivariate longitudinal measurements and survival data with applications to Parkinson's disease, *Statistical Methods in Medical Research*, 25(4), 1346-58, PMID: 23592717, PMCID: PMC3883896.
25. Li L, **Luo S**, Hu B, Greene T. (2017). Dynamic prediction of renal failure using longitudinal biomarkers in a cohort study of chronic kidney disease, *Statistics in Biosciences*, 9(2), 357-78, PMID: 29250207, PMCID: PMC5726783.
26. Wang J\*, **Luo S<sup>†</sup>**. (2017). Bayesian multivariate augmented Beta rectangular regression models for patient-reported outcomes and survival data, *Statistical Methods in Medical Research*, 26(4), 1684-99, PMID: 26037528, PMCID: PMC4457342.
27. Zhu H\*, **Luo S**, DeSantis, S. (2017). Zero-inflated count models for longitudinal measurements with heterogeneous random effects, *Statistical Methods in Medical Research*, 26(4), 1774-86, PMID: 26113383, PMCID: PMC5045322.

28. Su X\*, **Luo S**<sup>†</sup>. (2017). Analysis of censored longitudinal data with skewness and dependent censoring, *Communications in Statistics-Simulation and Computation*, 46(7), 5378-91, PMID: 29056818, PMCID: PMC5646848.
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## Letters

1. Chen Q, Zheng QW, **Luo, S.**, Zhao H. (2021). Comment on “Resection of the primary gastrointestinal neuroendocrine tumor improves survival with or without liver treatment”. *Annals of Surgery*, 274(6):e758-e759, PMID: 33065634, doi: 10.1097/SLA.0000000000004262.
2. Chen Y, **Luo, S.** (2011). A few remarks on “Statistical distribution of the difference of two proportions” by Nadarajah and Kotz. *Statistics in Medicine*, 30(15), 1913-1915. PMID: 21671250
3. Yi M., Meric-Bernstam F., Kuerer HM, Mittendorf EA, Bedrosian I, Lucci A, Hwang RF, Crow JR, **Luo S**, Hunt, K. (2012). Reply to K Van Zee, et al for comments on “Evaluation of a breast cancer nomogram for predicting risk of ipsilateral breast tumor recurrences in patients with ductal carcinoma in Situ after local excision”, *Journal of Clinical Oncology*, 30(25), 3144-5. doi: 10.1200/JCO.2012.43.9406.

Complete List of Published Work in MyBibliography:

<http://www.ncbi.nlm.nih.gov/sites/myncbi/1De4iwaVN475n/bibliography/44658186/public/?sort=date&direction=descending>

## Professional memberships

American Statistical Association (ASA)

International Chinese Statistical Association (ICSA)

The International Parkinson and Movement Disorder Society (MDS)

Institute of Mathematical Statistics (IMS)

American Association for the Advancement of Science (AAAS)

## **Mentoring and Education**

### **Short Course**

1. Design Clinical Trials with Non-inferiority Hypothesis: Methods and Applications, Society for Clinical Trials Annual Meeting, May 2022.
2. Design Clinical Trials with Multiple Endpoints: Methods and Applications, Society for Clinical Trials Annual Meeting, May 2021.

### **Faculty Mentoring Committee**

1. Dr. Zhiguo Li, Associate Professor of Biostatistics and Bioinformatics, Duke University.
2. Dr. Roland Matsouaka, Assistant Professor of Biostatistics and Bioinformatics, Duke University.
3. Dr. Hwanhee Hong, Assistant Professor of Biostatistics and Bioinformatics, Duke University.

### **Primary Postdoctoral Fellow Mentor**

1. Dr. Samsul Alam, PhD in Statistics from North Carolina State University, Jun 2024 - present.
2. Dr. Haotian Zhou, PhD in Biostatistics from University of North Carolina at Chapel Hill, Jun 2023 - present.
3. Dr. Dhrubajyoti Ghosh, PhD in Statistics from Washington University in St. Louis, Sep 2022 - August 2025, first position: Assistant Professor at Kennesaw State University.
4. Dr. Salil Koner, PhD in Statistics from North Carolina State University, Feb 2022 - June 2025, first position: Assistant Professor at University of California, Riverside.
5. Dr. Xiaoxiao Zhou, PhD in Statistics from The Chinese University of Hong Kong, March 2023 - Feb 2024, first position: Assistant Professor at University of Alabama at Birmingham.
6. Dr. Xiaoming Xu, PhD in Statistics from Colorado State University, July 2021 - Nov 2023, first position: AbbVie.
7. Dr. Yuanyuan Guo, PhD in Statistics from University of Missouri, July 2021 - Nov 2022.
8. Dr. Xiaofang Yan, PhD in Biostatistics from University of Louisville, Jan 2020 - Nov 2020, first position: Medpace.
9. Dr. Cai Li, PhD in Statistics from North Carolina State University, Jan 2019 - Sep 2019, first position: Assistant Member at St. Jude Children's Research Hospital.

### **Classroom Teaching**

#### **Department of Biostatistics & Bioinformatics, Duke University**

1. Categorical Data Analysis, Bio 714, Spring 2019, lead instructor, 100% responsibility.

#### **Department of Biostatistics, University of Texas School of Public Health**

1. Statistical Computing, PH 1998, Spring 2009, lead instructor, 100% responsibility.
2. Bayesian Data Analysis, PH 1998, Fall 2009, lead instructor, 100% responsibility.

3. Statistical Computing, PH 1998, Fall 2010, lead instructor, 100% responsibility.
4. Bayesian Data Analysis, PH 1998, Spring 2011, lead instructor, 100% responsibility.
5. Statistical Computing, PH 1998, Fall 2011, lead instructor, 100% responsibility.
6. Statistical Computing, PH 1930, Fall 2012, lead instructor, 100% responsibility.
7. Bayesian Data Analysis, PH 1965, Spring 2013, lead instructor, 100% responsibility.
8. Statistical Computing, PH 1930, Fall 2013, lead instructor, 100% responsibility.
9. Statistical Computing, PH 1930, Fall 2014, lead instructor, 100% responsibility.
10. Statistical Methods in Correlated Outcomes, PH 1918, Spring 2016, lead instructor, 100% responsibility.
11. Statistical Computing, PH 1930, Fall 2016, lead instructor, 100% responsibility.

#### **Primary PhD dissertation advisor**

1. Min Yi, doctoral in Biostatistics from UTHealth, graduated in Spring 2011, dissertation title: "Bayesian and survival model for tumor relapse status and disease-specific survival, with applications to breast cancer". First position: Senior Biostatistician at MD Anderson Cancer Center.
2. Geng Chen, doctoral in Biostatistics from UTHealth, graduated in Fall 2014, dissertation title: "Bayesian inference for multivariate longitudinal data analysis using robust distributions". First position: Senior biostatistician at GlaxoSmithKline.
3. Huirong Zhu, doctoral in Biostatistics from UTHealth, dissertation defense occurred on November 5, 2015, dissertation title: "Two-part mixture models for zero-inflated heterogeneous measurements and time to event data", co-advise with Stacia DeSantis, PhD. First position: Biostatistician at Texas Children Hospital.
4. Li-An Lin, doctoral in Biostatistics from UTHealth, dissertation defense occurred on November 19, 2015, dissertation title: "Bayesian analysis of multi-type recurrent events with dependent termination". First position: Senior biostatistician at Merck.
5. Jue Wang, doctoral in Biostatistics from UTHealth, dissertation defense occurred on May 19, 2016, dissertation title: "Dynamic predictions from joint models for multivariate longitudinal measurements and event time data". First position: Senior biostatistician at Genentech.
6. Ming Yang, doctoral in Biostatistics from UTHealth, dissertation defense occurred on November 29, 2016, dissertation title: "Bayesian quantile regression joint models: inference and dynamic predictions", co-advise with Stacia DeSantis, PhD. First position: Senior biostatistician at Genentech.
7. Kan Li, doctoral in Biostatistics from UTHealth, dissertation defense occurred on Feb 26, 2018, dissertation title: "Functional joint models for longitudinal and time-to-event data". First position: Senior biostatistician at Merck.
8. Jun Zhang, doctoral in Biostatistics from UTHealth, dissertation defense occurred on November 6, 2018, dissertation title: "Bayesian semi-parametric joint modeling for Parkinson's disease progression, competing failure times and missing data". First position: Senior biostatistician at Blue Cross Blue Shield.
9. Xuehan Ren, doctoral in Biostatistics from UTHealth, dissertation defense occurred on March 12, 2019, dissertation title: "Dynamic prediction of recurrent cardiovascular events". First position: Senior biostatistician at Gilead Sciences.
10. Jeffery Lin, doctoral study in Biostatistics, UTHealth, dissertation defense occurred on April 8, 2022, dissertation title: "Machine learning for the joint analysis of multivariate longitudinal and survival data". First position: Senior biostatistician at GlaxoSmithKline.

11. Haotian Zou, doctoral student in Biostatistics, The University of North Carolina at Chapel Hill, dissertation defense occurred on May 19, 2023, dissertation title: "Dynamic prediction with multivariate longitudinal outcomes and longitudinal magnetic resonance imaging data". First position: Postdoctoral fellow at Duke University.
12. Dongrak Choi, doctoral student in Biostatistics & Bioinformatics, Duke University, dissertation defense occurred on January 26, 2024, dissertation title: "An integrative modeling framework for neurodegenerative studies with longitudinal clinical data". First position: Senior biostatistician at Edwards Lifesciences Corp.

#### **PhD dissertation committee**

1. Lin Huo, doctoral in Biostatistics, UTHealth, graduated in Spring 2011, dissertation title: "Bayesian adaptive design for early phase drug-combination clinical trials".
2. Yunfei Wang, doctoral in Biostatistics, UTHealth, graduated in Spring 2012, dissertation title: "A simulation study using the hybrid statistics and the meta-analysis t-test for data with matched and unmatched subjects in the interim analysis of clinical trials".
3. Jin Young Byun, doctoral in Biostatistics, UTHealth, graduated in Spring 2012, dissertation title: "A hybrid method in combining treatment effects from matched and unmatched studies".
4. Rui Xia, doctoral in Biostatistics, UTHealth, graduated in year 2014, dissertation title: "A likelihood-based statistical method to detect DNA aberrations in SNP array data of tumor samples with very high normal cell contamination using haplotype structure derived from population genotypes".
5. Austin Brown, doctoral in Epidemiology, UTHealth, graduated in Spring 2014, dissertation title: "Healthcase professional weight loss advice, diet, physical activity and obesity in the fire service".
6. Julia Benoit, doctoral in Biostatistics, UTHealth, graduated in Spring 2014, dissertation title: "A hidden Markov model approach to analyze longitudinal ternary outcome subject to possible misclassification".
7. Michele Vidoni, doctoral student in Epidemiology, UTHealth, graduated in Spring 2016, dissertation title: "Vitamin B12 and homocysteine associations with physical function measures in older adults, data from the Baltimore longitudinal study of aging".
8. Yi-Ju Chiang, doctoral student in Biostatistics, UTHealth, dissertation proposal defense occurred on August 18, 2016, dissertation title: "Bivariate longitudinal logistic models with misclassified covariates and outcomes".

#### **Primary MS thesis advisor**

1. Eunji Jo, master student in Biostatistics, graduated in Fall 2009, thesis title: "Comparison of Bayesian and frequentist approaches in clinical outcomes between hospitals".
2. Minjeong Park, master student in Biostatistics, graduated in Spring 2010, thesis title: "Effects of air pollution on asthma hospitalization rates in different age groups in metropolitan cities of Korea".
3. Xiao Su, master in Biostatistics, graduated in May 2012, thesis title: "Bayesian estimation of multilevel item response model with missing data".
4. Gongbo Xie, master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2020, thesis title: "Model for Predicting Efficacy of Treatment and Complications in Women after Intradetrusor OnabotulinumtoxinA for Non-Neurogenic Urgency Incontinence".
5. Chun Xu, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2020, thesis title: "Physical function and accelerometer-based physical activity features in Physical Performance Across the Life-span Study (PALS)".

6. Huiyue Li, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2022, thesis title: "Novel gene variant in TP37, PI3R1, CALM1, and PLCG2 of the neurotrophin signaling pathway are associated with the progression from MCI to AD".
7. Caihan Wang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2022, thesis title: "Novel genetic variants in JAK2, CALM1 and PRKCA of the Biocarta biopeptide pathway are associated with progression from MCI to AD".
8. Songyun Zhang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2023, thesis title: "Assessing the necessity of regional nodal radiotherapy for breast cancer patients with positive ALN but pathologically regional lymph node negative disease after neoadjuvant chemotherapy".
9. Guanlin (Emery) Guo, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2024, thesis title: "Explainable Graph-Neural Network on neuroimaging-based cortical surface data to identify people with risk of Alzheimer's Disease".
10. Linke Li, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2024, thesis title: "biologically-explainable deep-learning based polygenic genetic risk score (PRS) for Alzheimer's disease".
11. Luowen Alison Yu, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2025, thesis title: "Longitudinal item response theory analysis of UHDRS: Tracking Huntingtons disease progression."
12. Yuxin Wang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2025, thesis title: "Comparing thymectomy and AZA/MMF regimens on time to MMS."
13. Tong Cheng, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2025, thesis title: "Race-specific machine learning models for stage IV breast cancer survival analysis."
14. Kexin Fang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2025, thesis title: "Comprehensive analysis of PIK3CA mutations and prognostic implications in advanced breast cancer: A secondary analysis of the SANDPIPER study."
15. Hanyu Wang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2025, thesis title: "Exploration of optimal synergistic strategies of postoperative radiotherapy and immunotherapy in early-stage breast cancer."
16. Nan Cen, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2025, thesis title: "Evaluating the impact of Azathioprine (AZA) and Mycophenolate Mofetil (MMF) on myasthenia gravis outcomes."
17. Yan Zhuang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2026, thesis title: "Axillary Management Strategies in Breast Cancer Patients with ypN0 after Neoadjuvant Therapy."
18. Bo Yang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2026, thesis title: "Comparing Cox Regression and Machine Learning for DLBCL Survival Prediction."
19. Junxin Zhang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2026, thesis title: "Overall Survival of Rare Histological Subtypes in Triple-Negative Breast Cancer Treated With Chemotherapy."
20. Shijun Wang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2026, thesis title: "Machine Learning-Based Overall Survival Prediction in Mixed Cellularity Classical Hodgkin Lymphoma: A NCDB Analysis."

**MS students thesis committee**

1. Meilin Jiang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2019, thesis title: "Data Visualization for Intensive Self-monitoring Glucose Trajectories of Type II Diabetes Patients and Predictive Modeling for Hemoglobin A1c Values".
2. Zhecheng Sheng, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2019, thesis title: "Analyzing the association of time-varying vital signs with in-hospital mortality & ICU transfer".
3. Meingdi Xuan, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2019, thesis title: "Comparing survival analysis methods in estimating treatment effect when treatment switching in a randomized clinical trial".
4. Farzana Noor Bindu, Master Student in Biostatistics, graduated in Spring 2012, thesis title: "Performance of classifying diabetes and diagnostic threshold for diabetes among the Mexican American border community".
5. Jin Li, Master Student in Biostatistics, graduated in Fall 2011, thesis title: "Nonlinear dimension reduction in pathway-based genome-wide association analysis".
6. Huandong Sun, Master Student in Biostatistics, graduated in Spring 2012, thesis title: "A computerized statistical framework for coalescent analysis".
7. Meilin Jiang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2019, thesis title: "Data Visualization for Intensive Self-monitoring Glucose Trajectories of Type II Diabetes Patients and Predictive Modeling for Hemoglobin A1c Values".
8. Zhecheng Sheng, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2019, thesis title: "Analyzing the association of time-varying vital signs with in-hospital mortality & ICU transfer".
9. Mengdi Xuan, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2019, thesis title: "Comparing survival analysis methods in estimating treatment effect when treatment switching in a randomized clinical trial".
10. Siqi Zhang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2020, thesis title: "Building a Prediction Model for Patient Enrollment in Clinical Trials Using Publicly Available Databases".
11. Chenyang Wang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2020, thesis title: "Bayesian Network Meta-Analysis of Trials Comparing Treatments for Schizophrenia".
12. Yijie He, 2nd year Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2021, thesis title: "Race Adjusted DNA damage repair SNP Analysis of C90401 Cohort for Prostate Cancer Association".
13. Blessing Ibe, 2nd year Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2022, thesis title: "Bayesian individual patient-level network meta-analysis of trials Comparing treatments for bipolar disorder".

#### **MPH students committee**

1. Naalia Sami, MPH in Biostatistics, graduated in Spring 2013, thesis title: "Effect of insurance status on stage of breast cancer diagnosis".
2. Simit Doshi, MPH graduate, thesis title: "Potential of role of statins as immunomodulators in treating pneumococcal pneumonia".
3. Jamie Marie Emert, MPH graduate, thesis title: "SEIS Community Needs Assessment Study for the Motherland and Hour Clinics in Houston, Texas".
4. Ashley A Francis, MPH graduate, thesis title: "Health significance of chlorination byproducts in drinking water: the Houston experience: a continuation study".



5. Oscar Sean Friendly, MPH graduate, thesis title: "Characterization of the east Houston particulate matter < 2.5 atmosphere: a novel method for desorbing organic constituents from airborne particulates".
6. Jacky T Lee, MPH graduate.
7. Alexander Charles Reese, MPH graduate, thesis title: "The effect of CATCH school lunch programs on obesity: a school based secondary analysis cross sectional study".
8. Yanneth Rivera, MPH graduate.

## Invited Talks, Presentations, and Posters

### Invited talks

1. Analysis of Smoking Cessation Patterns Using a Stochastic Mixed Effects Model with a Latent Cured State, *National Cancer Institute*, June, 2007.
2. Mixed Effects Stochastic Process Models of Smoking Cessation Behavior, *Department of Biostatistics, University of Texas at Houston*, November 2007.
3. Mixed Effects Stochastic Process Models of Smoking Cessation Behavior, *Department of Biostatistics, The University of North Carolina at Chapel Hill*, January 2008.
4. Mixed Effects Stochastic Process Models of Smoking Cessation Behavior, *Division of Biostatistics, Thomas Jefferson University*, January 2008.
5. Mixed Effects Stochastic Process Models of Smoking Cessation Behavior, *Department of Biostatistics, University of Florida*, February 2008.
6. Functional and Dynamic Data Analysis to Biomedical Research - Topic Contributed, Biometrics Section, *Joint Statistical Meetings (JSM)*, August 2009, Washington DC, session number: 254.
7. Classifying Ipsilateral Breast Tumor Relapse: A Novel Bayesian Approach, *Department of Biostatistics, University of Texas at Houston*, September 2009.
8. Binomial Regression with a Misclassified Covariate and Outcome, *Department of Biostatistics, University of Texas at Houston*, April 2010.
9. Center for AIDS Research and Analysis Core: Research Opportunities, *Baylor College of Medicine*, November 2010.
10. Methodological issues of using Item Response Models in Parkinson Disease Studies, *Department of Biostatistics, University of Texas at Houston*, April 2012.
11. Recent development of Multilevel Item Response Models with applications to Parkinson's Disease, *Department of Biostatistics, University of Texas at Houston*, January 2013.
12. The Use of Multilevel Item Response Models in Studying Parkinson's Disease, *Department of Mathematics, University of Texas at Arlington*, March 2013.
13. Robust Bayesian Inference for Multivariate Longitudinal Data Using Normal/Independent Distributions, *Houston Area Chapter, American Statistical Association*, February 2014.
14. Joint model of multivariate longitudinal data using multilevel item response model with applications to Parkinson's Disease, *The Western North American Region of the International Biometric Society Meeting (WNAR)*, June 2014, Honolulu, session number: WNAR Invited 1.
15. Bayesian Hierarchical Model for Multiple Repeated Measures and Survival Data: an application to Parkinson's Disease, *Third Joint Biostatistics Symposium*, June 2014, Chengdu, China.
16. Topic contributed: Joint Analysis of Multivariate Longitudinal Ordinal Measurements and Survival Data: An Application to Parkinson's Disease, *JSM*, August 2014, Boston, session number: 326.
17. Bayesian Analysis of Multi-type Recurrent Events and Dependent Termination with Nonparametric Covariate Functions, *Department of Preventive Medicine-Biostatistics, Northwestern University Feinberg School of Medicine*, October 6, 2014.

18. Bayesian Analysis of Multi-type Recurrent Events and Dependent Termination with Nonparametric Covariate Functions, *Department of Biostatistics & Epidemiology, University of Pennsylvania Perelman School of Medicine*, November 20, 2014.
19. Effect of Creatine Monohydrate on Clinical Progression in Patients With Parkinson Disease: A Randomized Clinical Trial, *Department of Biostatistics, University of Texas at Houston*, April 17, 2015.
20. Simultaneous inference of a misclassified outcome and competing risks failure time data, *Joint 24th ICSA Applied Statistics Symposium and 13th Graybill Conference*, June 2015, Fort Collins, Colorado, session number: 13.
21. Dynamic Prediction of Alzheimer's Disease Risk Based on Longitudinal Biomarkers and Functional Data, *JSM*, August 2016, Chicago, session number: 341.
22. Dynamic Prediction for Multiple Repeated Measures and Event Time Data: an Application to Parkinson's Disease, *The 10th ICSA International Conference*, December 2016, Shanghai, China, session number: 92.
23. Dynamic Prediction of Alzheimer's Disease Progression with Longitudinal Functional Joint Mode, *2017 Confence on Lifetime Data Science*, May 2017, Storrs, Connecticut, session number: 10.
24. Functional Joint Model with application of Alzheimer's disease, *2017 ICSA Applied Statistics Symposium*, June 2017, Chicago, session number: 78.
25. Dynamic Risk Prediction and Assessment, *Precision Medicine: Statistical Challenges and Opportunities Conference*, June 2017, Chicago, session number: 78.
26. Bayesian Functional Joint Models for Longitudinal and Time-To-Event Data, Department of Statistics, North Carolina State University, October 12, 2017.
27. Dynamic Risk Prediction and Assessment, Precision Medicine: Statistical Challenges and Opportunities Conference, May 8, 2018.
28. Dynamic Prediction of Alzheimer's Disease Progression: the Big Data Approach, Duke Clinical Research Institute, April 24, 2018.
29. Dynamic Risk Assessment and Prediction, *ASA/AdvaMed Medical Device Statistical Issues Conference*, May 8, 2018, Washington, DC.
30. Dynamic prediction using multivariate functional joint model: an application to Alzheimer's disease, *2018 International Chinese Statistical Association China Conference*, July 2018, Qingdao, China, session number: 6B.
31. Missing Data Issues in the Studies of Neurodegenerative Disorders: The Methodology, *JSM*, August 2018, Vancouver, session number: 44.
32. Dynamic Prediction of Alzheimer's Disease Progression Using Features of Multiple Longitudinal Outcomes and Time-to-Event Data, *The 2nd Conference on Lifetime Data Science*, May 2019, University of Pittsburgh, session number: 13.
33. Precision Medicine for Alzheimer's Disease Monitoring and Progression: the Dynamic Prediction Approach, *Division of Biostatistics, Washington University in St. Louis*, October 4, 2019.
34. Integrative Modeling and Dynamic Prediction of Alzheimer's Disease, *The Eastern North American Region of the International Biometric Society Spring Meeting (ENAR)*, Mar 2020, virtual meeting due to COVID-19, session number: 33.
35. Functional Data Analysis: Novel Statistical Methods and Applications in Medical Research, *Data-Driven Precision Medicine and Transnational Research in the Era of Big Data*, St. Jude Children Research Hospital, May 16, 2020.
36. Functional Data Analysis: Novel Statistical Methods and Applications in Alzheimer' Disease Research, *University of Nevada, Las Vegas*, April 28, 2022.
37. Functional Data Analysis: Novel Statistical Methods and Applications in Alzheimer' Disease Research, *The University of Mississippi Medical Center, Jackson, MS*, Sep 8, 2022.
38. Dynamic Prediction with Multivariate Longitudinal Outcomes and Longitudinal Magnetic Resonance Imaging Data, *ENAR*, Mar 2023, Nashville, session number: 13.

39. Deep Learning for the Dynamic Prediction of Multivariate Longitudinal and Survival Data, *JSM*, August 2023, Toronto, session number: 1172.
40. Integrative Modeling and Dynamic Prediction of Alzheimer's Disease, *University of Pittsburgh, Department of Biostatistics, Pittsburgh, PA*, Sep 7, 2023.
41. Integrative Modeling and Dynamic Prediction of Alzheimer's Disease, *North Carolina State University, Department of Statistics, Raleigh, NC*, Jan 16, 2024.
42. Advancing Alzheimer's Drug Development: A Novel Longitudinal Rank-Sum Test for Multiple Primary Endpoints in Randomized Controlled Trials, *ENAR*, Mar 2024, Baltimore, session number: 43.
43. Integrative Longitudinal Sparse Factor Modeling for Multi-Omics in an AI-Augmented Biomedical Landscape, *ENAR*, Mar 2026, Indianapolis, session number: 55.

### **Presentations and posters**

1. Contributed Talk: Modeling Individual Addiction Behavior Using a Mixed-Effect Model with Three States, *Joint Statistical Meetings (JSM)*, August 2006, Seattle, session number: 199.
2. Contributed Talk: Analysis of Smoking Cessation Patterns Using a Stochastic Mixed Effects Model with a Latent Cured State, *The Eastern North American Region of the International Biometric Society Spring Meeting (ENAR)*, March 2007, Atlanta, session number: 83.
3. Contributed Talk: Analysis of Smoking Cessation Patterns Using a Stochastic Mixed Effects Model with a Latent Cured State, *JSM*, August 2007, Salt Lake City, session number: 426.
4. Contributed Talk: Shrinkage Estimation for Robust and Efficient Screening of single SNP Association from Case-Control Genome-wide Association Studies, *ENAR*, March 2008, Arlington, session number: 45.
5. Contributed Talk: A Bayesian Approach for Correcting Misclassification in both Outcome Variable and Covariate, *ENAR*, March 2010, New Orleans, session number: 34.
6. Contributed Talk: Multivariate Longitudinal Bayesian Imputation for Global Statistical Test, with Applications to Parkinson's Disease, *JSM*, August 2010, Vancouver, Canada, session number: 128.
7. Contributed Talk: A Hybrid Method in Combining Treatment Effects from Matched and Unmatched Studies, presented by Jinyoung Byun\*, *JSM*, August 2010, Vancouver, Canada, session number: 631.
8. Contributed Talk: A Bayesian Model for Misclassified Binary Outcomes and Correlated Survival Data, with Applications to Breast Cancer, presented by Min Yi\*, *JSM*, August 2010, Vancouver, Canada, session number: 163.
9. Contributed Talk: Robust Bayesian Inference for Multivariate Longitudinal Data with Normal/Independent Distributions, *ENAR*, April 2012, Washington DC, session number: 78.
10. Poster Presentation: Analysis of Parkinson's Disease Data Using Item Response Model and Normal/Independent Distributions, *The International Parkinson and Movement Disorder Society's 16th International Congress*, June 2012, Dublin, Ireland, abstract number: 1297.
11. Contributed Talk: A Bayesian Item Response Model for Nonignorable Missing Data in Parkinson Disease Patients, *JSM*, July 2012, San Diego, session number: 244.
12. Contributed Talk: A Bayesian Approach to Joint Analysis of Parametric Accelerated Failure Time and Multivariate Longitudinal Data, *ENAR*, March 2013, Orlando, session number: 106.
13. Poster Presentation: Bayesian multiple imputation for missing multivariate longitudinal data from a Parkinson's disease clinical trial: a Power Analysis, *Society for Clinical Trials' 34th Annual Meeting*, May 2013, Boston.
14. Poster Presentation: Methods for Imputing Missing Data for Parkinson's Disease Clinical Trials with Multiple Correlated Outcomes, *The International Parkinson and Movement Disorder Society's 17th International Congress*, June 2013, Sydney, Australia, abstract number: 478.
15. Contributed Talk: Joint Modeling of Multivariate Longitudinal Measurements and Survival Data with Applications to Parkinson's Disease, *JSM*, August 2013, Montreal, Canada, session number: 308.

16. Session Chair: Developments in Genetic Association Studies, *JSM*, August 2013, Montreal, Canada, session number: 537.
17. Poster Presentation: Methods for Imputing Missing Data for Parkinson's Disease Clinical Trials with Multiple Correlated Outcomes, *Texas Regional Clinical and Translational Science Award (CTSA) Consortium Meeting*, November 2013, San Antonio.
18. Contributed Talk: Joint model for a diagnostic test without a gold standard in the presence of a dependent terminal event, *ENAR*, March 2014, Baltimore, session number: 72.
19. Session Chair: Imaging, *ENAR*, March 2014, Baltimore, session number: 58.
20. Session Chair: Recent Advances in Missing Data Methods, *The Western North American Region of the International Biometric Society Meeting (WNAIR)*, June 2014, Honolulu.
21. Session Chair: Methods for Current-Status and Interval-Censored Data, *JSM*, August 2015, Seattle, session number: 80.
22. Contributed Talk: Dynamic Predictions from Joint Models for Multivariate Longitudinal Measurements and Survival Data, *JSM*, August 2015, Seattle, session number: 608.
23. Poster Presentation: Dynamic prediction for multiple repeated measures and event time data: An application to Parkinson's disease, *The International Parkinson and Movement Disorder Society's 20th International Congress*, June 2016, Berlin, Germany, abstract number: 1869.
24. Contributed Talk: Dynamic Prediction of Alzheimer's Disease Risk Based on Longitudinal Biomarkers and Functional Data, *The 4th IMS Asian Pacific Rim Meeting*, June 2016, Hong Kong, session number: CP19.
25. Session Chair: Recent Challenges in -Omics Sciences, *JSM*, August 2016, Chicago, session number: 428.
26. Poster Presentation: Bayesian Analysis of Multiple Longitudinal Outcomes of Mixed Types with Nonparametric Treatment Effects, *ENAR*, March 2017, Washington DC.
27. Poster Presentation: Differential Item Functioning in the Unified Dyskinesia Rating Scale (UDysRS), *The International Parkinson and Movement Disorder Society's 21th International Congress*, June 2017, Vancouver, Canada, abstract number: 1141.
28. Poster Presentation: Missing Data in the Unified Dyskinesia Rating Scale (UDysRS), *The International Parkinson and Movement Disorder Society's 22nd International Congress*, Oct 2018, Hong Kong, China, abstract number: 1797.
29. Session Chair: ASA Biometrics Section JSM Travel Awards (I), *JSM*, July 2019, Denver, session number: 393.
30. Poster Presentation: Successful use of UDysRS regardless of PD- or Dyskinesia-duration, *The International Parkinson and Movement Disorder Society's 23rd International Congress*, Sep 2019, Nice, France, abstract number: 1190.